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Anomaly Detection of Breast Cancer Using Deep Learning

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Abstract
Cancer is one of the deadliest diseases facing humanity, one of the which is breast cancer, and it can be considered one of the primary causes of death for most women. Early detection and treatment can significantly improve outcomes and reduce the death rate and treatment costs. This article proposes an efficient and accurate deep learning-based anomaly detection framework. The framework aims to recognize breast abnormalities (benign and malignant) by considering normal data. Also, we address the problem of imbalanced data, which can be claimed to be a popular issue in the medical field. The framework consists of two stages: (1) data pre-processing (i.e., image pre-processing); and (2) feature extraction through the adoption of a MobileNetV2 pre-trained model. After that classification step, a single-layer perceptron is used. Two public datasets were used for the evaluation: INbreast and MIAS. The experimental results showed that the proposed framework is efficient and accurate in detecting anomalies (e.g., 81.40% to 97.36% in terms of area under the curve). As per the evaluation results, the proposed framework outperforms recent and relevant works and overcomes their limitations.

Keywords Anomaly detection · Breast cancer · Deep learning · Mammography · Diagnosis

Abbreviations

ACC	Accuracy
AD	Anomaly detection
AI	Artificial intelligence
AUC	Area under ROC curve
BI-RADS	Breast imaging-reporting and data system
CAE	Convolutional autoencoder
CBIS-DDSM	Curated breast imaging subset of the digital database for screening mammography
CC	Cranial-caudal
CNN	Convolutional neural network
CPU	Central processing unit
DBT	Digital breast tomosynthesis
DICOM	Digital imaging and communications in medicine
DL	Deep learning
F1	F1 score
FN	False negative
FNB	Fine needle biopsy
FP	False positive
FFDM	Full-field digital mammography
GAN	Generative adversarial network
GB	Gigabyte
GHz	Gigahertz
GPUs	Graphical processing units
MB	Megabyte
MIAS	Mammographic image analysis society
MLO	Mediolateral-oblique
MRI	Magnetic resonance imaging
MSE	Mean squared error
PGM	Portable gray map
ReLU	Rectified linear unit
ROC	Receiver operating characteristics
ROI	Region of interest
sec	Second
SENS	Sensitivity
SLP	Single-layer perceptron
SPEC	Specificity

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TN	True negative
TP	True positive
WHO	World Health Organization

1 Introduction

Cancer is one of the deadliest diseases facing humanity [1, 2]. Cells in almost any part of the body can become cancerous including the breast cells. Breast cancer is generated by the proliferation of cancerous cells in the breast and starts when breast cells begin to spiral out of control [1, 2]. Breast cancer becomes more common after the age of 40 [1, 3]. However, early detection and treatment can significantly improve outcomes, reduce the death rate and treatment cost [3, 4].

More women have died from breast cancer than any other disease, and thus, after lung cancer, it is the second most common cancer fatality in women [2, 5]. Based on the World Health Organization (WHO) statistics, over 2 million females with new breast cancer cases were diagnosed and 684,996 breast cancer deaths in 2020 in the world [6], as shown in Fig. 1. In Saudi Arabia, 3954 new breast cancer cases were diagnosed and 1095 breast cancer deaths in 2020 [7], as shown in Fig. 2. The 5-year prevalence in Saudi Arabia is 13,653 [7]. As per [8], by 2040, there will be 6886 more breast cancer cases than there are currently. (see Fig. 3).

There are some known challenges in diagnosing breast cancer as follows:

- Diagnosis techniques such as fine needle biopsy (FNB), in the breast cancer case, cause tumor rupture and accelerate the spread of cancer to the neighboring areas [2].
- Probably breast cancer can be detected after some symptoms appear, but sometimes many females with breast cancer exhibit no symptoms [9]. Due to this, breast cancer screening imaging techniques like mammography are important for diagnosing breast cancer.
- The mammologists may face a lot of breast mammography images, and it is required from them to diagnose all cases that they have in a short time, which may require effort and time; hence, such process can be error-prone.

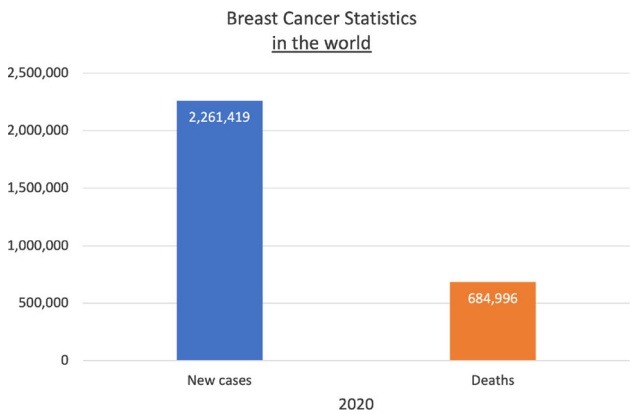


Fig. 1 Breast cancer statistics of the World Health Organization (WHO) in the world on 2020

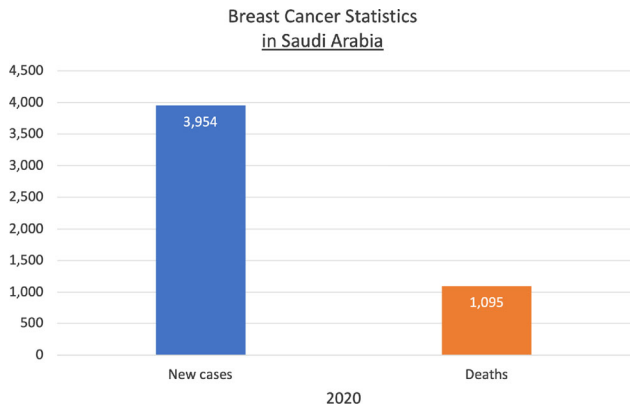


Fig. 2 Breast cancer statistics of the World Health Organization (WHO) in Saudi Arabia on 2020

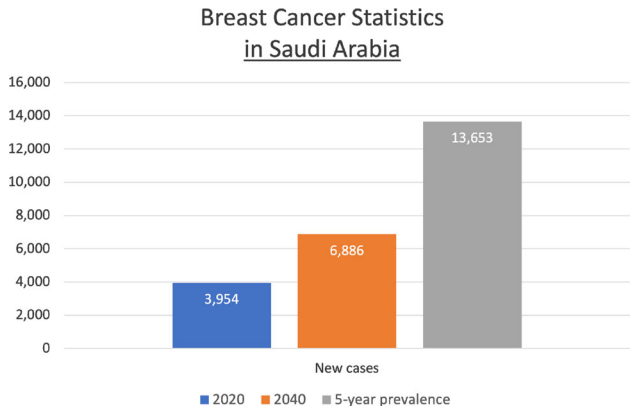


Fig. 3 Estimated past and future trends in total cases of breast cancer per year in Saudi Arabia

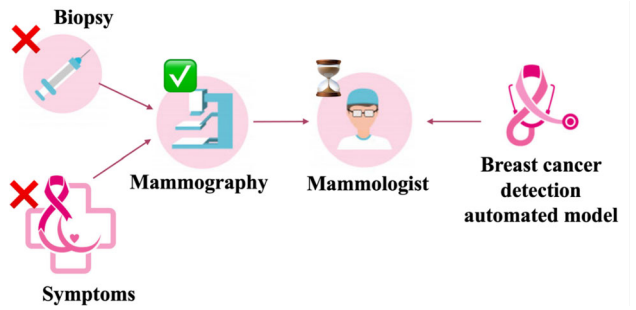


Fig. 4 Problem statement stages

Figure 4 presents the problem statement stages.

Further, employing artificial intelligence (AI) techniques in the medical field, in general, imposes several challenges.

One of these challenges is the lack of sufficient datasets. In the context breast cancer diagnose, and to the best of our knowledge, most of the previous works used a small dataset. Furthermore, to obtain the relevant labeling for these datasets represents another challenge. Thus, it would be difficult to recognize anomalies. Another data-related challenge is a class imbalance where the normal/healthy data is larger than the abnormal/disease data. Last but not least, the reliability of the algorithm used is vital, as such systems are considered critical (i.e., patients' life rely on them). Hence, it is important to consider the precision aspect of the algorithm used and its performance.

To the best of our knowledge, only few studies have used anomaly detection techniques for breast cancer. Most of the previous works solve this problem as a binary classification problem. Thus, it can be claimed that there is a lack in area of anomaly detection techniques for breast cancer diagnoses. Therefore, we believe that there is a need in the research community to adopt AI techniques (e.g., deep learning-based anomaly detection) in diagnosing breast cancer. This may avoid any error in diagnosis, reduce the number of breast cancer deaths and the challenges in the medical field mentioned above. Further, such research can contribute to the development of scientific researches in the field of anomaly detection techniques for breast cancer.

We propose an efficient and accurate anomaly detection framework (refer to Sect. 1) using deep learning to recognize breast abnormalities (benign and malignant) based on normal data through addressing the following research questions:

RQ1. How to develop an efficient and accurate anomaly detection framework for breast cancer diagnosis?
RQ2. Is the pre-processing of data will affect the performance of the proposed framework for breast cancer diagnosis?
RQ3. How to evaluate the effectiveness of the proposed framework for breast cancer diagnosis?

To evaluate the proposed framework, we conducted several experiments regarding the questions mentioned above. Also, we implemented our framework on two different datasets and compared the results. The experimental results showed that the proposed framework is efficient and accurate in detecting anomalies (e.g., 81.40-−97.36% in terms of area under the curve (AUC)). We also compared the performance of the proposed framework with recent and relevant works. As per the evaluation results, the framework proposed outperforms other recent and relevant works and overcomes their limitations (refer to Sect. 5).

We are motivated to work on this topic for a variety of reasons. First, to obtain an accurate and effective diagnosis during a short time, which is vital in processing medical images. This is because the manual processing of large data volumes is tedious, time-consuming, and impractical. Second, to increase the chance of survival and reduce the death rate caused by breast cancer. Third, to overcome the difficulty of obtaining a dataset containing all types of cancer cells and tissues, which we might need in the case of building binary classifiers [10]. Fourth, to improve the algorithm's performance to become more reliable (refer to Sect. 6). Finally, to help mammologists and preserve their time and effort.

The major contribution and innovations of this study are as follows:

1. An efficient and accurate anomaly detection framework using deep learning to detect breast cancer (malignant only) and breast abnormalities (benign and malignant) as an anomaly from normal data using two different datasets.
2. Optimizes the performance of the proposed framework to overcome the performance of recent and relevant works.
3. Identifies the most suitable data pre-processing pipeline for each dataset used and demonstrates the effect of these pre-processing pipelines on the performance of the proposed framework.
4. Evaluates the effectiveness of the proposed framework using different evaluation metrics and compares them with other recent and relevant works.

The rest of this paper is organized as follows: In Sect. 2, we provide the essential background to understand the rest of the paper. Section 3 introduces the recent works of this study. It lists previous works in the same field and presents how our framework differs from previous ones. Section 4 introduces the conceptual framework of the proposed framework. It presents the stages and steps of the framework in detail. Section 5 presents the experiments and the proposed framework results with two different datasets. Also, it presents multiple and various experiments to answer the questions of this study. Section 6 discusses the outcome of the proposed framework. Section 7 illustrates the limitations of this work. Finally, the conclusion and future work are in Sect. 8.

2 Background

The background information needed to understand the rest of the paper is explained in this section. We provide a succinct explanation of the following terminologies: anomaly detection, deep learning, deep learning-based anomaly detection, and automated medical image diagnosis in detail.

2.1 Anomaly Detection (AD)

2.1.1 What is Anomaly Detection?

The technique of locating data instances that differ greatly from other data instances is known as anomaly detection, also referred to as outlier detection [11, 12]. Also, anomaly detection is defined as identifying samples that differ from the usual or predicted [12, 13]. The considerations of the anomaly are various according to the field. In the images, e.g., white tigers between regular tigers may be considered as an anomaly [14]. Also, breast cancer among normal breast may be considered an anomaly. Also, viral pneumonia or COVID-19 virus among non-viral pneumonia and healthy controls may be considered an anomaly.

2.1.2 Types of Anomaly

Anomalies are classified into three types: point anomalies, contextual anomalies and collective anomalies.

1. **Point Anomalies** A point anomalies occur when a single instance in a dataset that varies from others in terms of its attributes [15]. The majority of studies in the literature focus on point anomalies [14]. This is the type used in this thesis.
2. **Contextual Anomalies** The contextual anomalies are also known as conditional anomalies, and they occur when the data is abnormal in some specific context [14, 15].
3. **Collective Anomalies** Collective anomalies, also known as group anomalies, occur when a group of associated data instances is anomalous to the overall data set. The single data instances in a collective anomaly may not be anomalies by themselves, but it is unusual for them to occur as a group [14, 16].

2.1.3 Input of Anomaly Detection Techniques

In deep anomaly detection methods, a deep neural network architecture is primarily determined by the nature of the input data. There are two types of input data: sequential (e.g., voice, text, music, time series, protein sequences) and non-sequential data (e.g., images, other data) [14]. The input data type of this thesis is images which are non-sequential data.

2.1.4 Output of Anomaly Detection Techniques

The outcomes of anomaly detection methods are classified into two types: anomaly score or binary labels.

1. **Anomaly Score** "The outlierness level for each data point" is described by an anomaly score. Anomaly scores generally reveal more information than binary labels. [14].
2. **Binary Labels** Rather than assigning scores, some techniques may label each data instance as normal or anomaly /abnormal [14].

2.2 Deep Learning (DL)

Due to the widespread of deep neural networks, which produced exceptional results in a wide range of applications, many researchers tended to use deep learning to detect abnormalities in images. By inspecting every pixel in an image, it can also deal with complex features [11, 12, 14]. Furthermore, deep learning has shown promising success due to a large amount of data and the high computational power of Graphical Processing Units (GPUs) [12, 17].

2.3 Deep Learning-Based Anomaly Detection

Recently, deep learning-based anomaly detection attempts to automatically learn features, e.g., using Convolutional Neural Networks (CNNs) in vision [12, 18, 19]. Furthermore, deep learning-based anomaly detection overcomes the issue of imbalanced data which is a popular issue in the medical field. Since the positive cases (e.g. cancer cases) are very few and difficult availabilities compared to negative cases (e.g. normal cases) [12].

2.4 Automated Medical Image Diagnosis

2.4.1 What is Diagnostic Imaging?

Diagnostic imaging refers to a variety of techniques used to examine the inside of the body in order to determine the causes of an illness or injury and confirm a diagnosis. It also helps to determine how well your body reacts to disease or fracture treatment. Some machines and devices can produce images of your body's activities and structures [20, 21].

2.4.2 Types of Diagnostic Imaging

The following are the most popular types of diagnostic imaging.

- **Mammography** Mammography is a technique of X-ray imaging used to diagnose and screen the human breast [22]. It is one of the most reliable screening tools, as well as an early detection tool because it can detect breast cancer at an early stage before physical symptoms appear [1, 5, 17]. Mammography is widely used because it is widely available and inexpensive. Mammograms are obtained from two different views for each breast: cranial-caudal

(CC) and mediolateral-oblique (MLO) (see Fig. 5) [5, 23].

- **X-Rays** X-rays are one of the most well-known and widely used diagnostic imaging tests. The x-ray machine generates a high-energy beam that cannot be absorbed by bones or dense tissue but passes through the rest of the body. Then it generates an image of any part of the body that shows if there is any damage to it [20, 21, 24].
- **Magnetic Resonance Imaging (MRI)** Magnetic Resonance Imaging, or MRI, is a non-invasive imaging technique that produces three-dimensional detailed anatomical images. It is used to detect illnesses, make a diagnosis, and monitor patient response to treatment [24, 25].
- **CT Scans** Computed Tomography (CT) is made up of a series of X-ray images taken from various angles. Then, the computer software produces cross-section (slices) images of the body. It is better than standard X-rays because it provides a more efficient picture [20, 21, 24].
- **Ultrasound** Ultrasounds imaging techniques generate images of the inside of the body safely. High-frequency waves are used instead of radiation. Ultrasound images show the movement of internal organs and blood flow through the vessels in real-time [20, 21].

Figure 6 illustrates some examples of diagnostic imaging types.

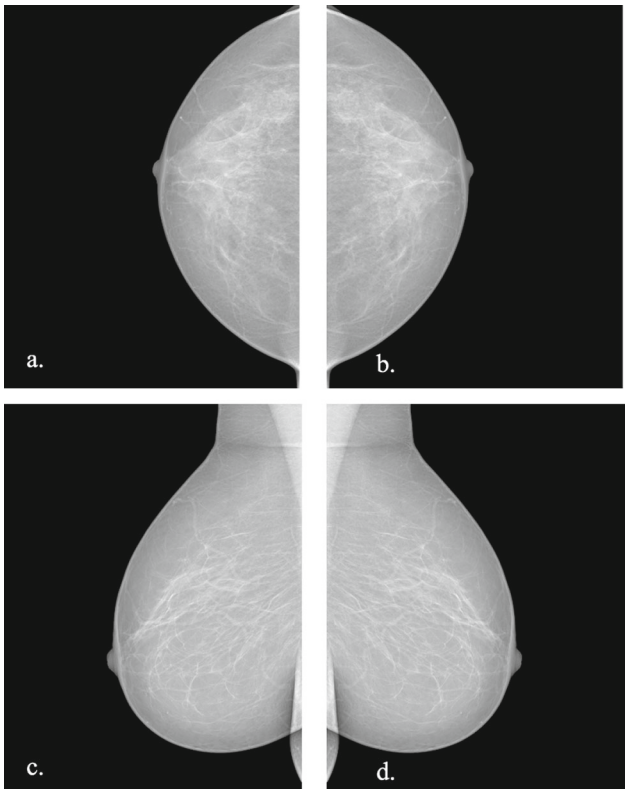


Fig. 5 Mammogram examples taken from INbreast dataset [23]: **a** cranial-caudal (CC) view of the right breast; **b** CC view of the left breast; **c** mediolateral-oblique (MLO) view of the right breast; **d** MLO view of the left breast

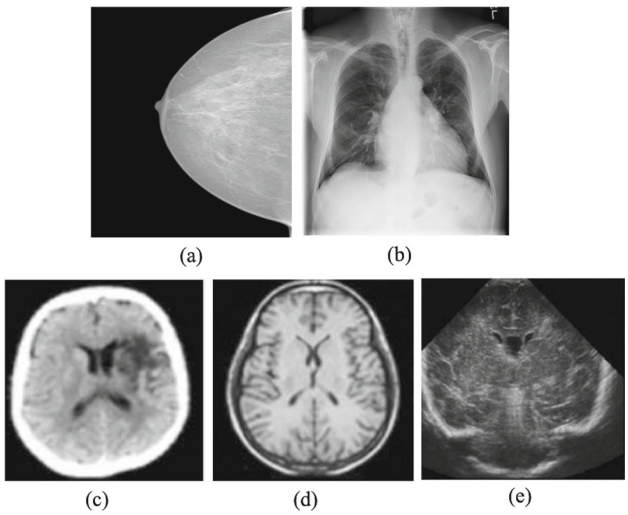


Fig. 6 Some examples of diagnostic imaging types: **a** Mammography breast from INbreast dataset [23]; **b** chest X-rays from X-ray dataset [26]; **c** CT scan; **d** MRI; **e** Ultrasound [27] of the same brain image

3 Related Work

This section reviews and summarizes different previous studies about deep learning-based anomaly detection mechanisms for detecting breast cancer.

Wei et al. [28] developed a deep neural network-based autoencoder to detect anomalies in medical images based one-class classification. It used INbreast dataset which is a public dataset. In terms of Area Under ROC Curve (AUC), the performance was 84%. However, still a small dataset was employed in this study. They did not assess their outcomes against recent research or state-of-the-art techniques.

Ahmadi et al. [10] proposed an architecture for irregular tissues in mammography images using two deep convolutional networks (R and M). They trained their method using two public datasets: the INbreast and the Mammographic Image Analysis Society (MIAS) datasets. To test their approach, they used the Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS-DDSM) dataset. In terms of AUC, they achieved 86% and 76% in INbreast and the MIAS datasets, respectively. But, the datasets used are small. Furthermore, they did not consider the whole image processing in a single phase.

Gao and Wu [29] proposed a novel approach, which is a new measure for locating the impact of a specific sample on a task, enabling them to detect samples that are not in the distribution. For the anomaly recognition task, their approach was incorporated into a simple convolutional autoencoder (CAE) model. The private datasets Breast Full-Field Digital Mammography (FFDM) and Breast Magnetic Resonance Imaging (MRI) were used to test their methodology. According to the results of the experiment, the proposed approach outperforms the compared methods with 90.1% and 95.6% in terms of AUC in MRI and FFDM datasets, respectively. The method's experiments were conducted on relatively small datasets.

In Fujioka et al. [30], they used generative adversarial network (GAN)-based anomaly detection on breast ultrasound images. The model detects abnormal lesions in images and calculates the anomaly score as abnormalities. Breast ultrasound images were obtained from Tokyo Medical and Dental University Hospital. The dataset contains 531 normal images from 69 patients, with data augmentation resulting in 6372 normal images for training. The experimental results show that they achieved high performance in detecting and diagnosing anomalous lesions on breast ultrasound images, especially in differentiating between normal tissues from malignant masses. However, they did not contrast their findings with state-of-the-art techniques or other works.

According to new research by Swiecicki et al. [31], they proposed a generative adversarial network-based abnormality detection using only normal images to detect breast cancer. The model is being trained for use in digital breast tomosynthesis. The digital breast tomosynthesis (DBT) dataset was used which is a 3D Mammography. It is made up of 19,230 reconstructed volumes from 4348 different patients. They study two types of cases: normal and cancer cases. They used mean squared error (MSE) and discriminator losses as measures in the experiments. The results indicate that cancer locations had a significantly higher completion error than non-cancer locations (mean error ratio of 2.77). However, they did not contrast their outcomes with state-of-the-art techniques or other works.

See this article [12] for a further comprehensive analysis of previous works in the area of image anomaly detection using deep learning in general and in the medical field in particular.

Table 1 summarizes the most prominent differences between the works mentioned above and our proposed framework in terms of methodology, dataset, pre-processing, results and limitations.

4 Conceptual Framework

This section presents the proposed framework that detects breast cancer using a deep learning-based anomaly detection technique in detail. Section 4.1 gives the overall description of the proposed framework. Sections 4.1.1 and 4.1.2 introduce more details about the proposed framework.

4.1 The Proposed Framework

The proposed anomaly detection framework detects breast abnormal either malignant only (cancer) or benign and malignant (abnormal) by training normal data. In this work, there are two main sub-models implemented on two different datasets: INbreast dataset [23] and Mammographic Image Analysis Society (MIAS) dataset [32] (see Figs. 7 and 8). The proposed framework mainly consists of two major stages, as shown in Fig. 7. The first stage is the data pre-processing stage, in which the images are pre-processed with appropriate steps to each dataset as shown in Figs. 7 and 8. After that, the output from the first stage, which is the pre-processed images, is used as input to the second stage to extract the deep features using the Convolutional Neural Network (CNN). Then the classifier network takes the features extracted by the feature extractor network and predicts whether the sample belongs to the normal class, cancer class, or abnormal class.

4.1.1 Data Pre-processing Stage

Pre-processing of data including augmentation techniques is an essential step to obtain good performance. Therefore, the data pre-processing is the first stage of the proposed framework, as shown in stage 1 in Figs. 7 and 8. Two different datasets were used. Each of them has different pre-processing steps according to the dataset properties.

1. *INbreast Dataset.* In the INbreast dataset [23], there are five pre-processing steps (see stage 1 in Fig. 7). First, extracting Region of Interest (ROI). Second, cropping images. Third, applying contrast stretching by contrast stretching for each image based on the following Eq. 1:

$$x_2 = 255 \times \frac{x_1 - \min}{\max - \min} \tag{1}$$

where x_1, x_2 denote old gray level and new gray level, respectively. min, max denote minimum and maximum gray values for each image, respectively. After that, the augmentation technique and feature scaling are applied as the fourth and fifth steps. For the augmentation step, the horizontal and vertical shift augmentation techniques [33] were implemented. It occurs when all of the pixels in the image are moved in one direction horizontally or ver-

Table 1 Related work summary

[Ref.] (Year)	Methodology	Dataset (#: sample size)	Pre-processing	Results (%:performance of the model used)	Finding/limitation
Wei et al. [28] (2018)	A deep neural network-based autoencoder. Feature extraction and classification: autoencoder model	INbreast dataset: (410)	NA	AUC: Normal versus Malignant: 84%	1. Used small datasets. 2. No pre-processing data 3. No process the whole image in one step 4. No comparison with other algorithms
Ahmadi et al. [10] (2019)	An adversarial training architecture based on two deep convolutional networks (R and M). Feature extraction: encoder–decoder networks as Reconstruction Network (R). Classification: involves convolution and fully connected layers as Matching Network (M)	1. Mammographic image analysis society (MIAS) dataset: (322) 2. INbreast dataset: (410)	NA	AUC: MIAS: Normal versus Benign and Malignant: 76% INbreast: Normal versus Benign & Malignant: 86%	1. Small datasets were used 2. There is no pre-processing data 3. No process the entire image in single phase
Gao and Wu [29](2020)	A new measure for locating the impact of a specific sample on a task, enabling to detect samples outside of distribution. Feature extraction and classification: convolutional autoencoder (CAE model)	1. Breast magnetic resonance imaging (MRI): (2872) 2. Breast full-field digital mammography (FFDM): (304)	Resize of image	AUC: 1. MRI: 90.1% 2. FFDM: 95.6%	Small datasets were used
Fujioka et al. [30] (2020)	GAN-based anomaly detection on breast ultrasound images. Feature extraction and classification: generative adversarial network (GAN)	Breast ultrasound images: (531 normal images for training). Data augmentation: (6372 normal images for training)	Image resizing	AUC: 1. Normal versus Benign + malignant: 93.6% 2. Normal versus Malignant: 98.5% 3. Normal versus Benign: 86.2% 4. Normal + Benign versus Malignant: 86.3%	There is no comparison with state-of-the-art methods or other works
Swiecicki et al. [31] (2021)	A generative adversarial network-based abnormality detection using only normal images for model training with application to digital breast tomosynthesis. Feature extraction and classification: generative adversarial network (GAN)	Digital breast tomosynthesis (DBT)(3D mammography): (4348 patients)	1. Extract patches from images. 2. Covering each patch with a square-shaped mask of pre-determined side	Mean error ratio: 2.77	There is no comparison with other works or state-of-the-art methods

Table 1 continued

[Ref.] (Year)	Methodology	Dataset (#: sample size)	Pre-processing	Results (%:performance of the model used)	Finding/limitation
This work	A deep learning-based anomaly detection mechanism for detecting breast cancer. **Feature extraction:** CNN: MobileNetV2 network. **Classification:** Single-layer perceptron (SLP) network.	1. INbreast dataset: (4640–6360) 2. Mammographic image analysis society (MIAS) dataset: (5160–6440)	**INbreast dataset:** 1. Extracting Region of Interest (ROI). 2. Cropping images. 3. Applying contrast stretching 4. Augmentation technique. 5. Feature scaling. **MIAS dataset:** 1. Augmentation technique. 2. Feature scaling.	AUC: **INbreast dataset:** 1. Normal versus Malignant: 94.25% 2. Normal versus Benign & Malignant: 89.79% **MIAS dataset:** 1. Normal versus Malignant: 97.36% 2. Normal versus Benign & Malignant: 81.40%	Overcome all the above limitations by introducing: 1. An efficient and accurate anomaly detection framework for breast cancer diagnosis 2. An efficient and accurate anomaly detection framework for breast abnormal diagnosis 3. Using two different datasets (INbreast and MIAS datasets) 4. Selecting appropriate pre-processing steps for each dataset to have the best results 5. Obtaining remarkable performance and outperforms the compared other relevant works 6. Overcome the limitation of previous works in [10, 28] 7. Retraining and testing the two main sub-models without pre-processing steps in both datasets to illustrate the effect of these pre-processing steps on the performance of the proposed framework 8. Answering all research questions with prominent results

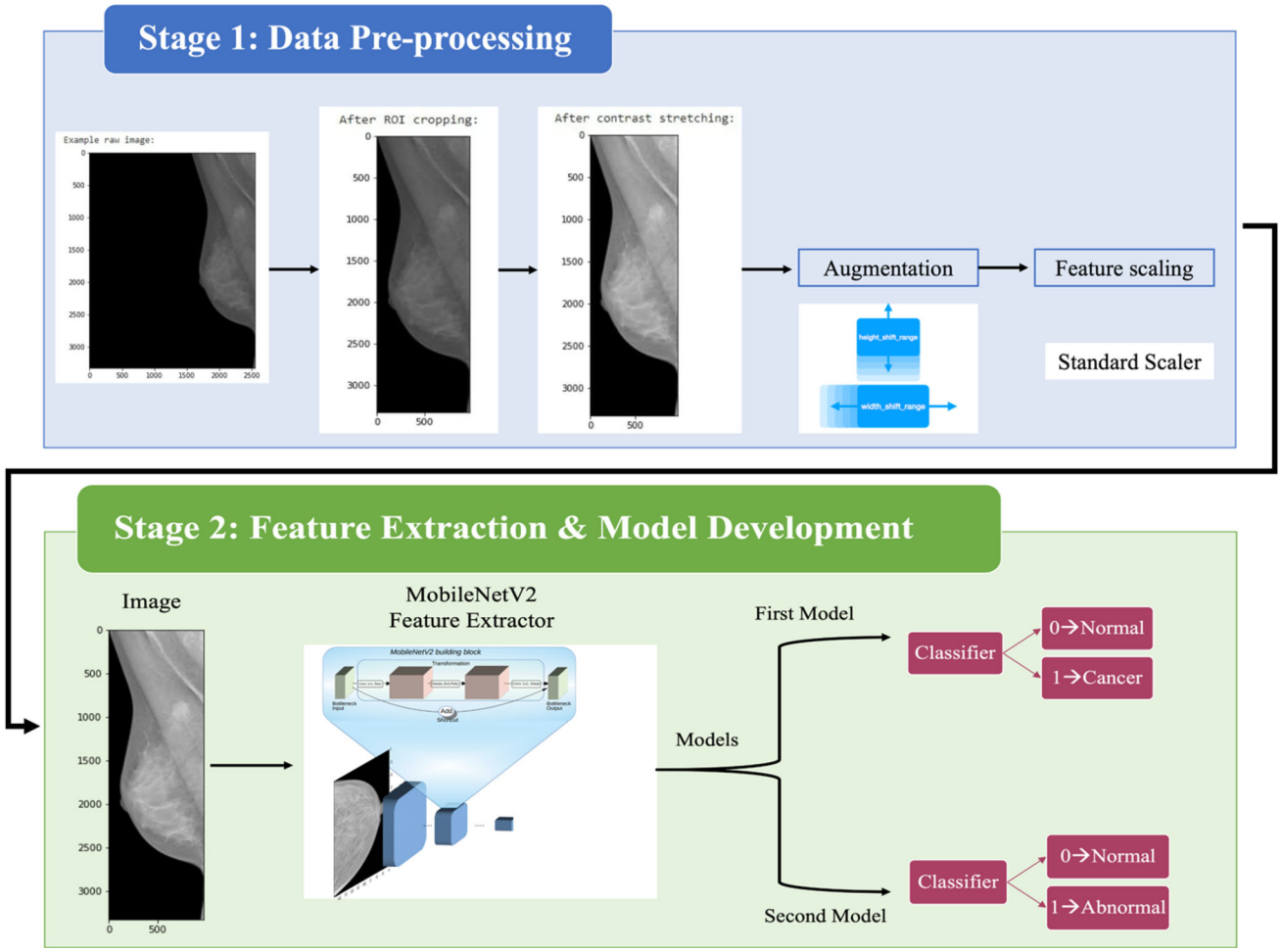


Fig. 7 Stages of the proposed model for INbreast dataset

tically, however with maintaining the image's dimensions [34]. For the feature scaling step, the Standard Scaler method was used based on the following Eq. 2:

$$z = \frac{x - \mu}{\sigma} \tag{2}$$

where z denote standard scores of the samples (also called z scores). μ denotes mean and σ denotes standard deviation.

2. *MIAS Dataset.* In the MIAS dataset [32], there are two pre-processing steps (see stage 1 in Fig. 8). First, the augmentation step whihc is the same technique used in INbreast dataset as mentioned above. Second, feature scaling, which also used the Standard Scaler method as in INbreast dataset. It was difficult to extract Region of Interest (ROI) in this dataset because the shape of the images makes it hard to extract ROI without cutting any part of the breast. Furthermore, the images of this dataset already roughly focus on the breast region, not like the INbreast dataset. The contrast stretching was not applied in this dataset because the data already has good and enough contrast.

All these pre-processing steps were selected carefully and implemented to enhance and achieve the best possible results.

4.1.2 Feature Extraction and Model Development Stage

This section introduces the second stage of the proposed framework, as shown in stage 2 in Figs. 7 and 8. The included sub-items explain the feature extraction network using a deep learning approach. Moreover, it presents how to generate fake data images using the Gaussian noise. The extracted features are then used to build classification models using the single-layer perceptron (SLP) classifier.

1. *Feature Extraction* The second stage of the proposed framework begins with feature extraction. The generated pre-processed images from the previous stage are used as input to the pre-trained Convolutional Neural Net-

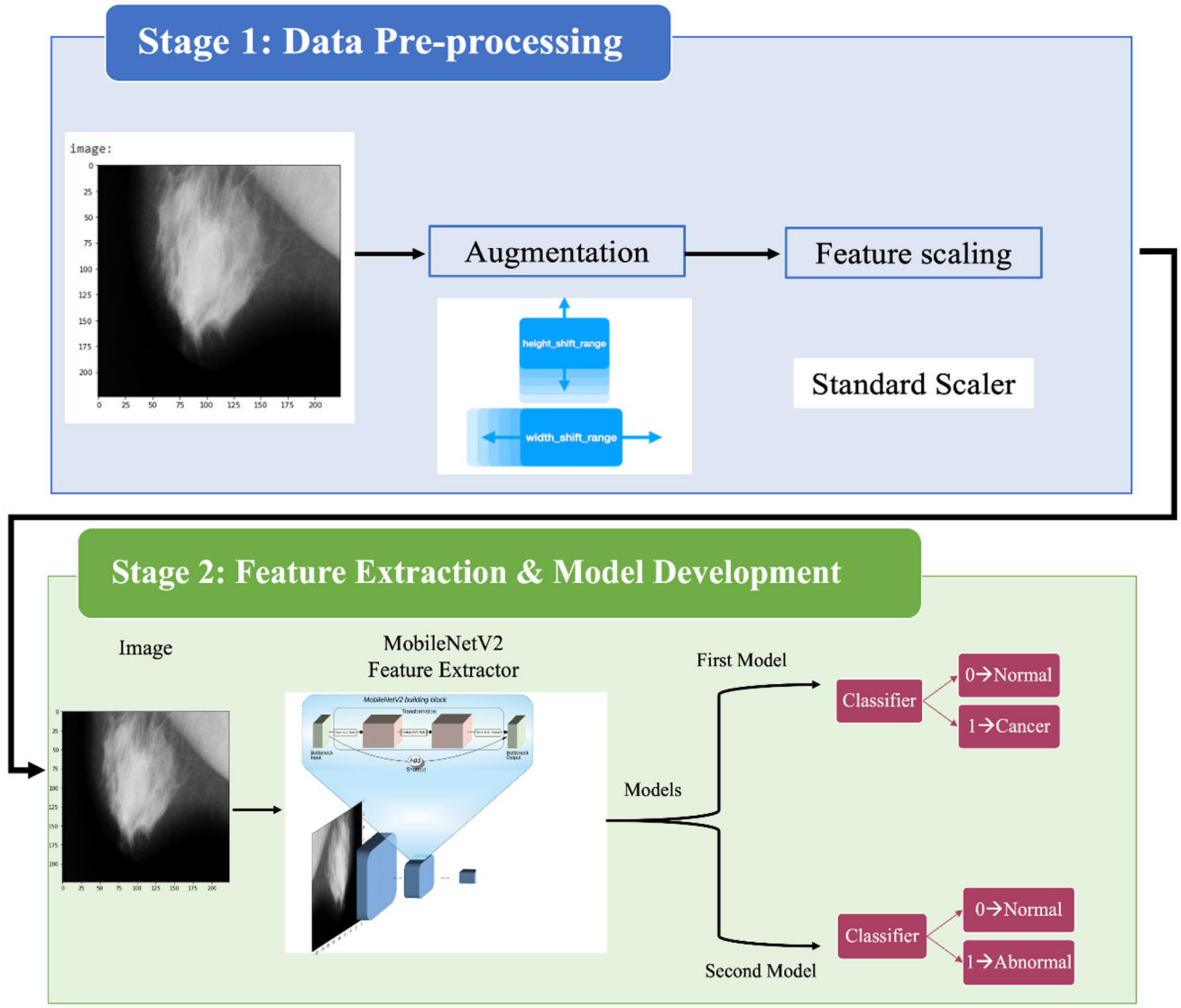


Fig. 8 Stages of the proposed model for MIAS dataset

work (CNN) 'MobileNetV2' [35] to extract features. The strength and main feature of CNN is that it converted the manual feature extraction process into an automated process. It gives better image recognition when neural network feature extraction becomes deeper (contains more layers) [36]. The MobileNetV2 pre-trained model was used for some reasons. First, it works well on mobile devices [35]. Second, since it can perform well on personal mobile devices, it will improve user experience and energy consumption [37]. Third, it is the smallest pre-trained model with a size of 14 MB [38]. Fourth, it has 3.4 million parameters [35] and its main advantage is that the number of parameters is reduced in the neural network, which results in lightweight deep neural networks [39]. Fifth, the most important thing is that the MobileNetV2 is faster in performance [35, 37]. Finally, it is a very effective feature extractor for object detection with all these features, and the best possible results have been achieved. In MobileNetV2, there are two kinds of blocks. The first is a 1-stride residual block. The other is the block with a stride of 2, which is used for downsizing [40]. Each has three layers. The first layer is 1×1 convolution with a Rectified Linear Unit (ReLU6). The depthwise convolution is the second layer. The third layer is another nonlinear 1×1 convolution [40] (see Figs. 9 and 10). Meanwhile, the MobileNetV2 network architecture consists of 19 bottleneck residual blocks starting with the initial fully convolution layer with 32 filters, as shown in Table 2. Then after the residual bottleneck blocks, the 1×1 convolution layer with an average pooling layer. A classification layer is the last layer [35, 41]. It requires an input of size $224 \times 224 \times 3$, where 3 is the num-

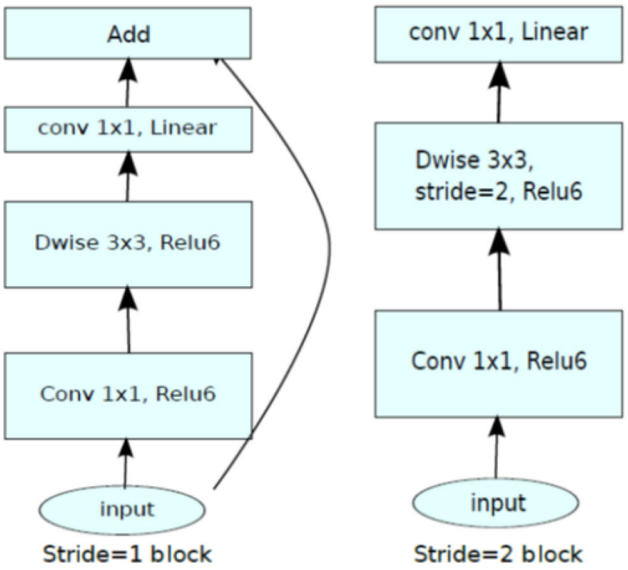


Fig. 9 The MobileNetV2 convolutional blocks [35]

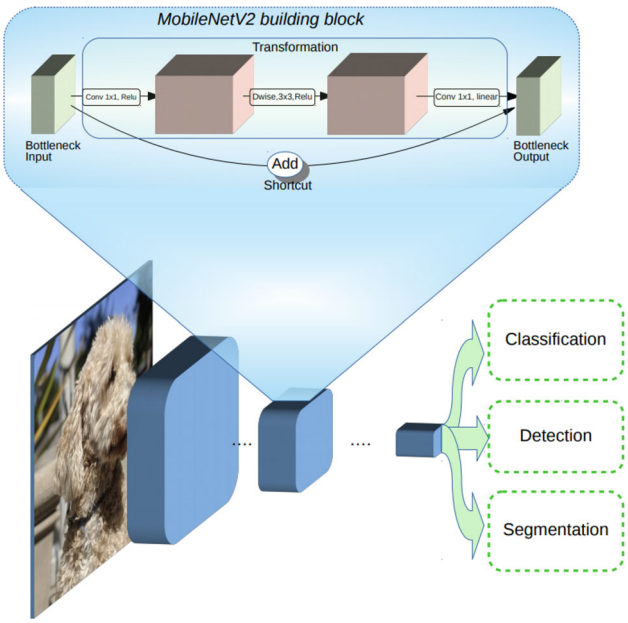


Fig. 10 MobileNetV2 architecture overview. Composite convolutional building blocks are represented by blue blocks [37]

ber of color channels. Batch normalization is used during training [35].

As mentioned above, the MobileNetV2 network is used for feature extraction only. So, the last classification layer is not needed in this phase. Therefore, the include_top argument flag in MobileNetV2 pre-trained was setting to False (include_top=False), and hence the classification layers at the top of the MobileNetV2 network are omitted, which is ideal for feature extraction. In addition, before compiling and training the model, the convolutional base was frozen

Table 2 MobileNetV2 architecture [35]

Input	Operator	t	c	n	s
$224^2 \times 3$	conv2d	–	32	1	2
$112^2 \times 32$	Bottleneck	1	16	1	1
$112^2 \times 16$	Bottleneck	6	24	2	2
$56^2 \times 24$	Bottleneck	6	32	3	2
$28^2 \times 32$	Bottleneck	6	64	4	2
$14^2 \times 64$	Bottleneck	6	96	3	1
$14^2 \times 96$	Bottleneck	6	160	3	2
$7^2 \times 160$	Bottleneck	6	320	1	1
$7^2 \times 320$	Conv2d 1×1	–	1280	1	1
$7^2 \times 1280$	Avgpool 7×7	–	–	1	–
$1 \times 1 \times 1280$	Conv2d 1×1	–	k	–	

t: The factor of expansion, c: The number of output channels, n: repeating number, s: stride. 3×3 kernels are used for spatial convolution

to prevent weights in a given layer from being updated during training [42]. It was frozen by setting the trainable flag of the model to False (pretrained_model.trainable = False). Finally, the MobileNetV2 network extracted 62720 features from each image for all input images.

- *Features for Fake Data Images (Gaussian Noise):* We generate (same number of normal samples) fake data samples for training by taking the features extracted from the normal training samples and adding Gaussian noise to them. The Gaussian noise has the same mean and standard deviation as those of the features of the normal training samples. The steps of the proposed framework for both two main sub-models in both two datasets are illustrated in Figs. 11 and 12. First, the dataset was split into train, validation, and test. Then, it was passed to the MobileNetV2 feature extractor to extract features. After that, we got the training features of normal and added the Gaussian noise to them with the same mean and standard deviation of the normal training samples features to generate fake data. In the end, we added the fake data features in training beside the original features of normal to help the model during training. Then, training the model and validation using features of validation samples for both two types, normal and cancer (Fig. 11) or normal and abnormal in the second sub-model (Fig. 12). Finally, testing the model by test samples and getting the results. This process was implemented step by step.

2. *Model Development* Model development is the process of building a model using the features extracted from the dataset. This work selected a single-layer perceptron (SLP) network with a sigmoid activation function as a classifier network. The perceptron takes as input

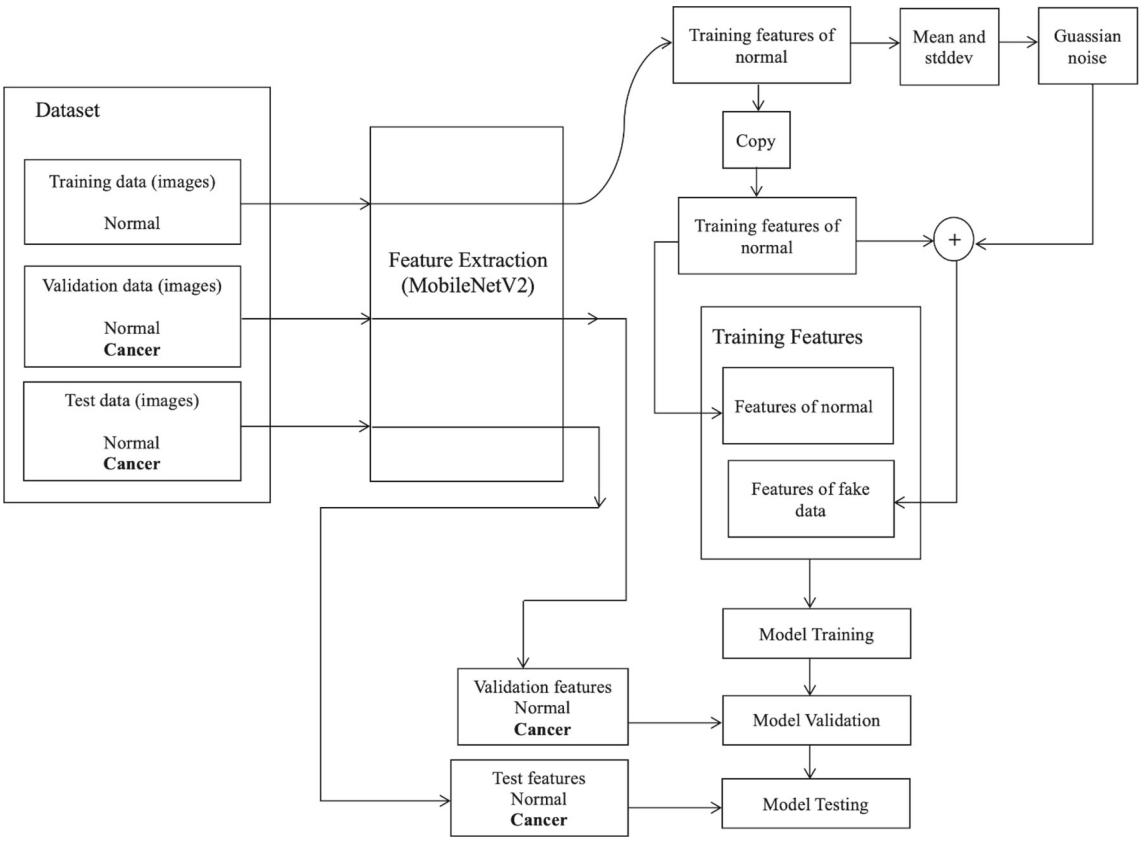


Fig. 11 Steps of the Normal versus Cancer (Malignant only) model in both two datasets (step by step process)

the 62720 features extracted by MobileNetV2 and then classifies the input. The classification network's output is either 1 or 0. In this case, 1 represents a data sample from the target class, and 0 represents a data sample from the negative class. The SLP has achieved the best accuracy results. In this research, two main sub-models were implemented. The first model is designed to classify the data into normal class or cancer (malignant only) class. Here, 0 corresponds to the normal class, and 1 corresponds to the cancer class. The second model is designed to classify the data into normal class or abnormal (benign and malignant) class. Also, in this sub-model, 0 corresponds to the normal class, and 1 corresponds to the abnormal class (see stage 2 in Figs. 7 and 8).

5 Experiments and Results

This section introduces the experimentation setup details in Sect. 5.1. Also, it presents a discussion of the experimental results in Sect. 5.2 to evaluate all techniques used and answer the questions of this research.

5.1 Experimentation Setup

The following subsections present the datasets description, training settings, evaluation metrics, and development tools and hardware.

5.1.1 Datasets Description

Two different datasets are used in this work. Both of them are public datasets. The datasets are: INbreast dataset [23] and MIAS dataset [32]. Each of these datasets is described in detail in the following subsections.

1. ***INbreast Dataset.*** This dataset [23] is public dataset, and it includes 410 mammography images with a resolution of 3000 × 4000 in mediolateral-oblique (MLO) and cranial-caudal (CC) views. This dataset has several types of lesions, including benign and malignant cancer. The images are saved in Digital Imaging and Communications in Medicine (DICOM) format. Two cases are considered in this work. First, we considered only cancer cases (malignant) as anomalies versus normal cases. Second, we considered abnormal cases (benign

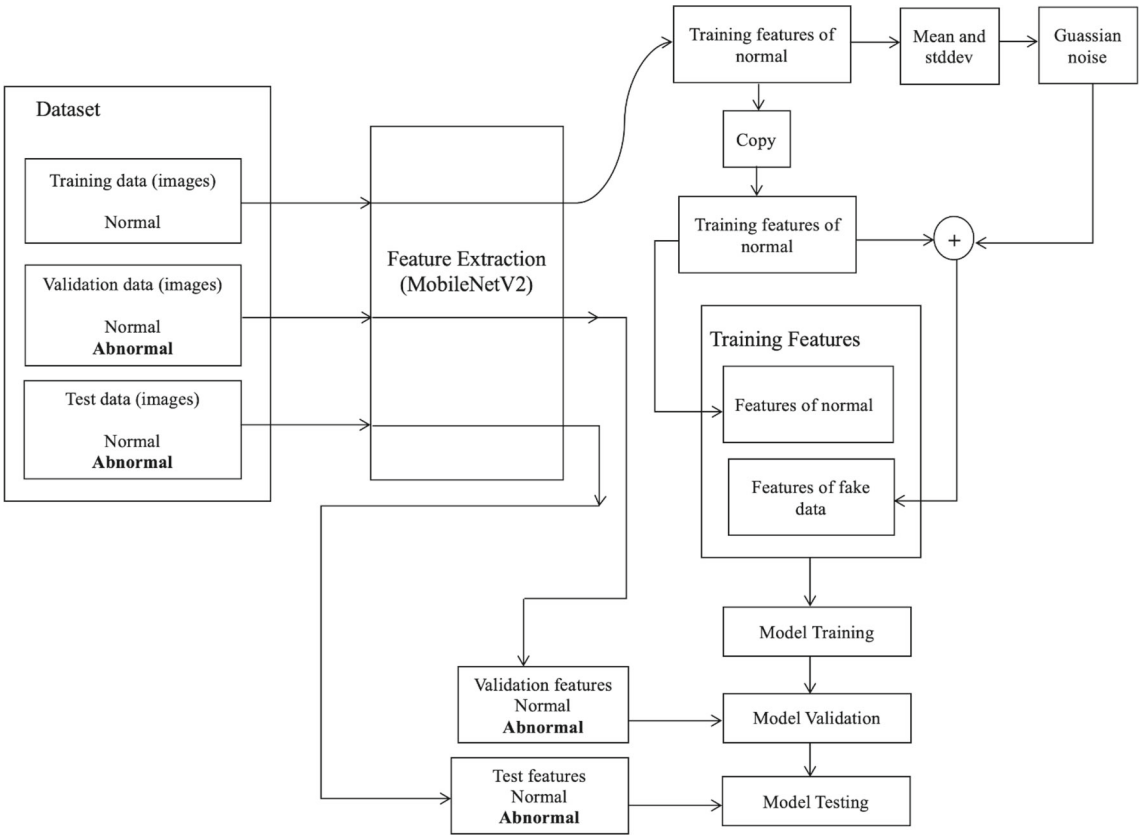


Fig. 12 Steps of the Normal versus Abnormal (Benign and Malignant) model in both two datasets (step by step process)

and malignant) as anomalies versus normal cases. Table 3 shows the details of the INbreast dataset.

2. ***MIAS Dataset.*** This dataset [32] is a public, and it contains 322 mammography images in mediolateral-oblique (MLO) view only with a 1024 × 1024 resolution. This dataset has several types of lesions, including benign and malignant cancer. The images are saved in Portable Gray Map (PGM) format. Also, two cases are considered in this study. First, we considered only cancer cases (malignant) as anomalies versus normal cases. Second, we considered abnormal cases (benign and malignant) as anomalies versus normal cases. Table 4 shows the details of the MIAS dataset.

As mentioned in (Sects. 4 and Section 4.1.1), the augmentation technique was used in the pre-processing data stage. Therefore, this subsection illustrates the number of images used in the proposed framework with and without augmentation (original images) in both datasets: INbreast and MIAS datasets and in both sub-models as shown in Table 5.

5.1.2 Training Settings

The proposed framework was trained using the two datasets described in the previous section. The dataset was split into train, validation and test sets-77% for training, 12% for validation and 11% for testing. The training is on negative data only plus fake data from negative data (see Sect. 4). In addition, 5% of positive data is training besides negative data. All these processes will help the model during training and will improve the model's performance. The positive data has been used as a part of the validation set to ensure the model can distinguish between negative and positive data as well as it used in the testing set. The number of positive data is set to equal to the number of negative data that rest from the training set, even though a number of positive data is much more than the negative data since not used all in training. Therefore, this setting will achieve data balance and avoid bias the model to positive data in validation and testing sets.

The framework has been trained using Adam optimization algorithm with a learning rate of 0.001, a batch size of 32, with epochs of 60. The input shape is (224, 224, 3) where input size 224 × 224 and 3 is the number of color channels. The loss function that is used to train the framework is binary

Table 3 INbreast dataset details

Origin	Year	Image size	Number of cases	Number of images	Views	Image type file	BI-RADS
Portugal	2010	3328 × 4084 OR 2560 × 3328 pixels	115	410	MLO, CC	DICOM	Yes

Origin	Year	BI-RADS used	Number of normal images	Number of benign images	Number of malignant images	Dataset type
Portugal	2010	1 → normal 4 → benign 5 → Malignant	67	43	49	Public

MLO mediolateral-oblique view, *CC* cranial-caudal view, *DICOM* digital imaging and communications in medicine; *BI-RADS* breast imaging-reporting and data system

cross-entropy. Table 6 summarizes the list of parameters used in the proposed framework.

5.1.3 Evaluation Metrics

The proposed frameworkon several and various performance metrics where this section will present it. As well as we computed the model training time, training time average per epoch, and average predicting time per sample. The evaluation metrics are illustrated in detail as follows:

- **Accuracy (ACC):** The accuracy [43] is the ratio of correctly classified samples to the total number of samples in the dataset. This metric is not very efficient in the anomaly detection approach; in other words, it is not very useful when dealing with unbalanced datasets [44]. However, still an important and common metric. The accuracy Eq. 3:

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{3}$$

- **F1 Score (F1):** The F1 score [43] is "the harmonic mean of precision and recall (also known as sensitivity)". It gives good results on imbalanced datasets [45]. The F1 score Eq. 4:

$$F1 = 2 \times \frac{\mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}} = \frac{2 \times \mathrm{TP}}{2 \times \mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{4}$$

- **Area Under ROC Curve (AUC):** The Area Under the ROC curve (AUC) [46] indicates how well the model's performance can distinguish between two classes. It uses true positive rates (TPR/Sensitivity (SENS)) and false positive rates (FPR /1-Specificity (SPEC)). The higher the AUC, the better the model distinguishes between patients with and without the disease. This metric is popular and commonly used with the anomaly detection approach because it gives effective results [47]. The equations used in AUC and ROC Curve are as follows:

$$\mathrm{SENS} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{5}$$

$$\mathrm{SPEC} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \tag{6}$$

$$\mathrm{FPR} = 1 - \mathrm{SPEC} = \frac{\mathrm{FP}}{\mathrm{TN} + \mathrm{FP}} \tag{7}$$

- **Training Time (s):** The training time of the models was computed in seconds.
- **Training Time Average per Epoch (s):** Also, the average training time per epoch of the models was computed in seconds.
- **Average Predicting Time per Sample (s):** The average needed time for predicting the sample was computed in seconds.
- **Some Testing Examples for Predictions from the Model:** The models have been tested using some unseen samples in the same dataset and views their predictions. It displays the samples with predictions. This is not a metric.

The proposed framework was implemented using Python programming language in Colaboratory (Colab) environment. The Colab is a notebook-based code editor which uses Ipython (an interactive shell built with python). The Python version used is 3.7.12. The framework was run on NVIDIA Tesla K80 graphics processor, with Intel®Xeon®CPU @ 2.30 GHz processor, and a memory of 13 GB.

5.2 Results

This section provides the results of several experiments of the proposed framework to answer the research questions. We have three research questions.

Table 4 MIAS dataset details

Origin	Year	Image size	Number of cases	Number of images	Views	Image type file	BI-RADS
UK	1994	1024 × 1024 pixels	161	322	MLO	PGM	No

Origin	Year	Number of normal images	Number of benign images	Number of malignant images	Dataset type
UK	1994	207	64	51	Public

UK United Kingdom, *MLO* mediolateral-oblique view, *PGM* portable gray map, *BI-RADS* breast imaging-reporting and data system

Table 5 Number of images used in the proposed framework

Dataset	Augmentation	Negative	Positive	Model
INbreast dataset	No	67	49	Normal versus Cancer
	Yes (40 copies)	2680	1960	
	No	67	92	Normal versus Abnormal
	Yes (40 copies)	2680	3680	
MIAS dataset	No	207	51	Normal versus Cancer
	Yes (20 copies)	4140	1020	
	No	207	115	Normal versus Abnormal
	Yes (20 copies)	4140	2300	

Table 6 Results of Normal versus Cancer (Malignant only) and Normal versus Abnormal (Benign and Malignant) models with data pre-processing stage in INbreast dataset

Model	AUC	F1 Score	Accuracy	Training time total (s)	Training time avg per epoch (s)	Avg predicting time per sample (s)
Normal versus Cancer	94.25%	91.14%	91.84%	82.97	1.38	0.39
Normal versus Abnormal	89.79%	87.17%	88.44%	82.78	1.38	0.42

RQ1. How to develop an efficient and accurate anomaly detection framework for breast cancer diagnosis?
RQ2. Is the pre-processing of data will affect the performance of the proposed framework for breast cancer diagnosis?
RQ3. How to evaluate the effectiveness of the proposed framework for breast cancer diagnosis?

The experiments will be structured as follows (see Fig. 13).

5.2.1 Experiment 1: RQ1. How to Develop an Efficient and Accurate Anomaly Detection Framework for Breast Cancer Diagnosis?

The answer to this question includes two parts. The first part of the answer to the question of **how to develop anomaly detection framework** is by combining pre-processing steps, feature extraction, and anomaly detection module (for more information, see Sect. 4). The results efficiency and accuracy of the framework are proven, which are presented in this section, and it is the second part of the answer to the question of **how to develop an efficient and accurate anomaly detection framework** for the first research question.

- ***With Data Pre-processing Stage***

1. **INbreast Dataset.** This experiment evaluate the results of the Normal versus Cancer (Malignant only) model and the Normal versus Abnormal (Benign and Malignant) model in the INbreast dataset [23]. The models were trained on normal data, and they recognized breast cancer and breast abnormalities from normal data. Table 6 shows the results of both models in terms of AUC, F1 score, accuracy, and time complexity. The AUC metric is the suitable metric to measure the performance of anomaly detection (imbalanced datasets), which will be used to measure the efficiency and accuracy of the model's performance. The Normal versus Cancer model achieved 94.25% in terms of AUC, 91.14% in F1 score, and 91.84% in accuracy. While the Normal versus Abnormal model achieved 89.79% in terms of AUC, 87.17% in F1 score, and 88.44% in accuracy (see Table 6).

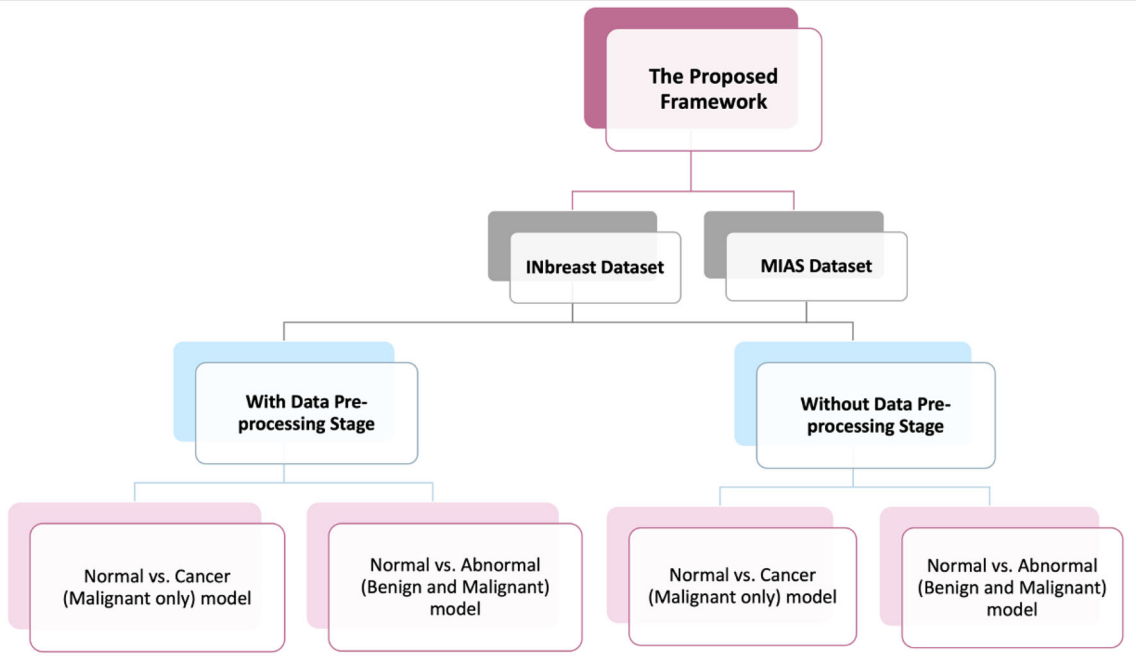


Fig. 13 The experiments taxonomy of the proposed framework

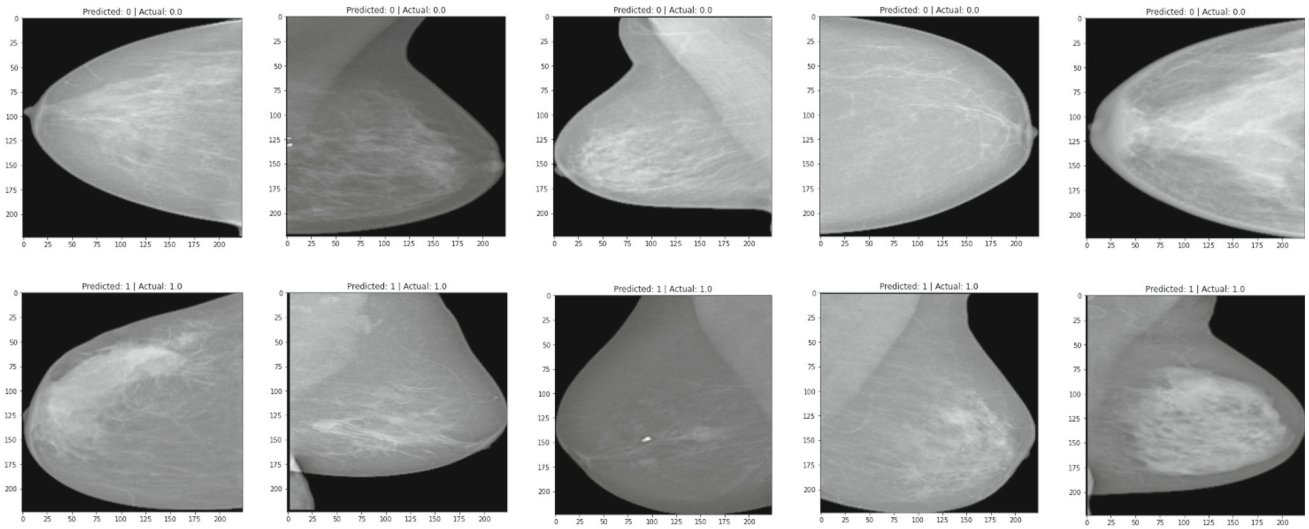


Fig. 14 Some testing examples from the Normal versus Cancer (Malignant only) model with data pre-processing stage in INbreast dataset. where 1 = positive (cancer), 0 = negative (normal)

Figures 14 and 15 present the predictions of some testing examples from both models. In the Normal versus Cancer model, there are ten examples and they all predicted correctly. In the Normal versus Abnormal model, there is one example predicted wrong from ten examples. The wrong example is from positive samples.

2. **MIAS Dataset.** This experiment evaluates the results of the Normal versus Cancer (Malignant only) model and the Normal versus Abnormal (Benign and Malignant) model in the MIAS dataset [32]. Table 7 shows the results of both models in terms of AUC, F1 score, accuracy, and time complexity.

The AUC of the Normal versus Cancer model is 97.36%, The F1 score is 92.15%, and the accuracy is 91.76%, while the AUC of Normal versus Abnormal model is 81.40%, F1 score is 72.59%, and the accuracy is 75.60% (see Table 7).

Figures 16 and 17 present the predictions of some testing examples from both models. There are ten

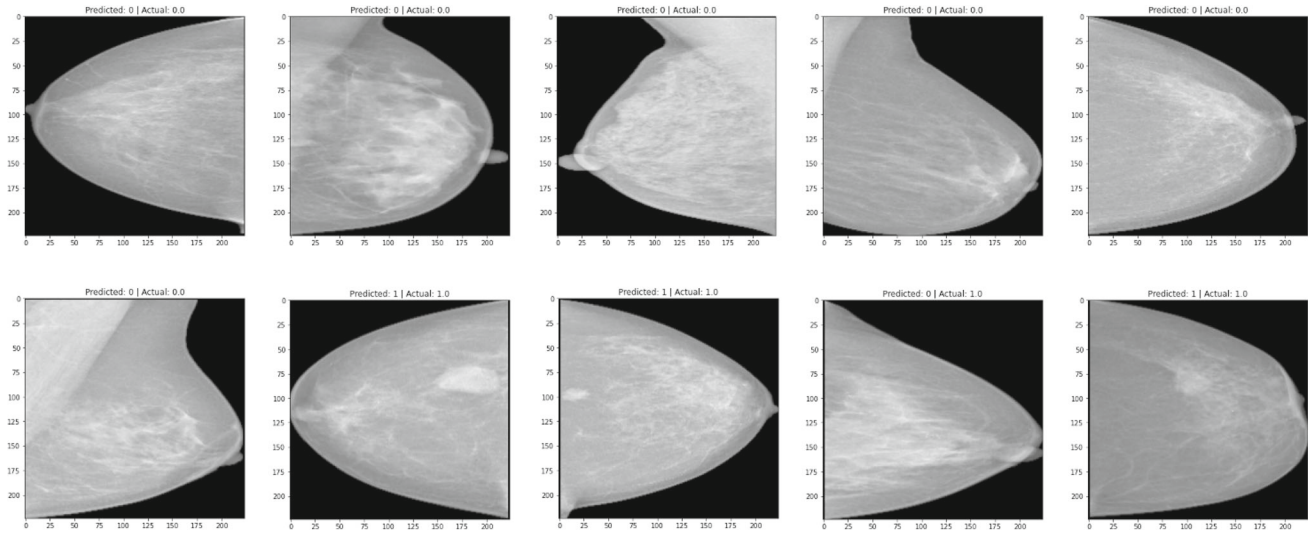


Fig. 15 Some testing examples from the Normal versus Abnormal (Benign and Malignant) model with data pre-processing stage in INbreast dataset, where 1 = positive (abnormal), 0 = negative (normal)

Table 7 Results of Normal versus Cancer (Malignant only) and Normal versus Abnormal (Benign and Malignant) models with data pre-processing stage in MIAS dataset

Model	AUC	F1 Score	Accuracy	Training time total (s)	Training time avg per epoch (s)	Avg predicting time per sample (s)
Normal versus Cancer	97.36%	92.15%	91.76%	80.25	1.34	0.57
Normal versus Abnormal	81.40%	72.59%	75.60%	83.26	1.39	0.56

examples, and each one is correctly predicted in Normal versus Cancer model, while in Normal versus Abnormal model, from ten examples, one was incorrectly predicted. The incorrect example comes from positive samples.

In this section, we answered the first question of the research questions with outstanding performance for the framework. However, we faced challenges in both datasets which are difficult to deal with and the work with them needs to be accurate. Since the features are not very clear and the images are similar, there is no obvious difference between them.

5.2.2 Experiment 2: RQ2. Is the Pre-processing of Data Will Affect the Performance of the Proposed Framework for Breast Cancer Diagnosis?

To answer the second question of the research questions, the data pre-processing stage was removed, keeping the second stage and all settings the same. Then the proposed framework was re-experimented to see whether the pre-processing of data would affect the proposed framework's performance and, in addition, whether the effect is huge or not.

- ***Without Data Pre-processing Stage***

1. **INbreast Dataset.** The previous section presents the major experiments of this study, in which the models have the data pre-processing steps. This experiment evaluates the results of the Normal versus Cancer (Malignant only) model and the Normal versus Abnormal (Benign and Malignant) model without data pre-processing steps in the INbreast dataset [23]. It recognized breast cancer and breast abnormalities from normal data without any pre-processing data steps. Table 8 shows the results of the both models with and without data pre-processing stage in terms of AUC, F1 score, accuracy, and time complexity.
We notice from Table 8 that both models results are significantly decreased. The difference ratio between the Normal versus Cancer models with and without data pre-processing of AUC is 16.7%, the F1 score is 19.71%, and accuracy 20.41%. While the difference ratio between the Normal versus Abnormal models with and without data pre-processing of AUC is 24.49%, the F1 score is 24.67%, and accuracy 31.3%.
Figures 18 and 19 present the predictions of some testing examples from both models. In Normal versus Cancer model, we can see that there are three examples from ten examples are mispredicted. One

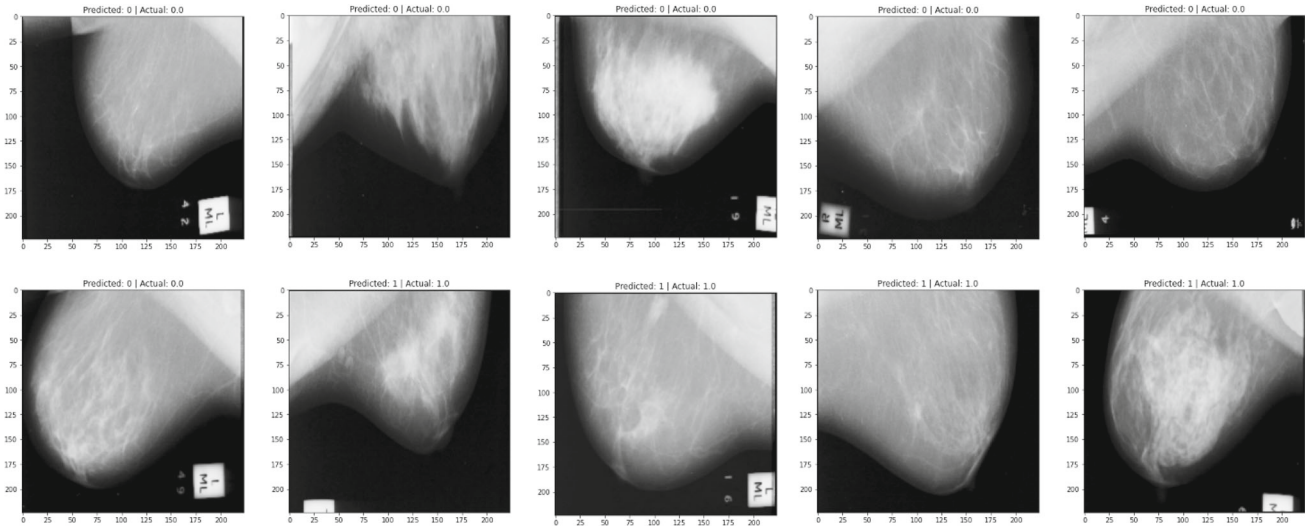


Fig. 16 Some testing examples from the Normal versus Cancer (Malignant only) model with data pre-processing stage in MIAS dataset, where 1 = positive (cancer), 0 = negative (normal)

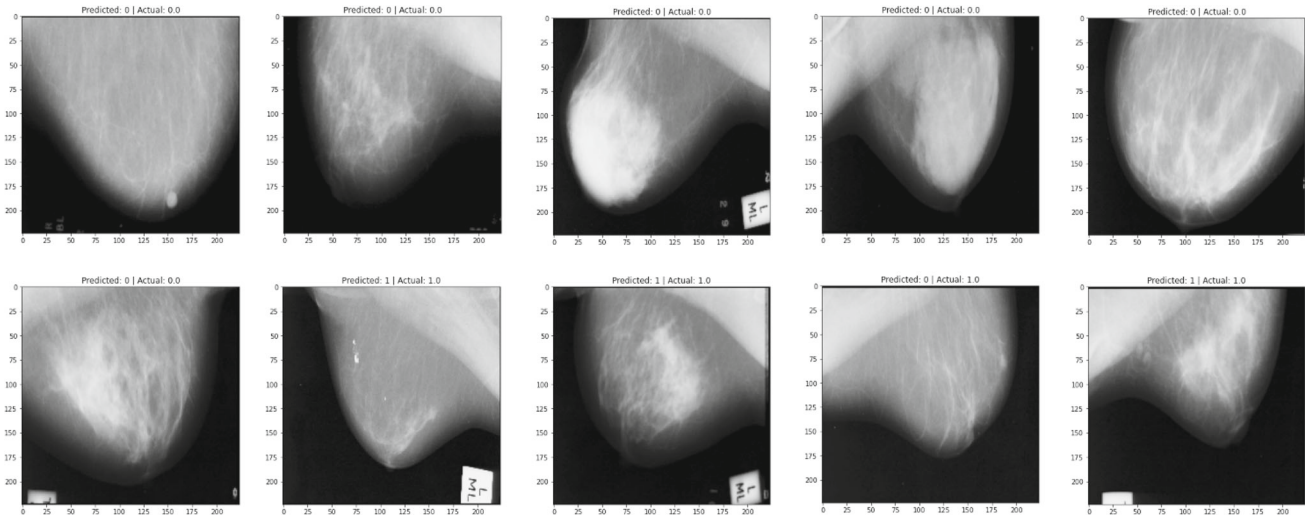


Fig. 17 Some testing examples from the Normal versus Abnormal (Benign and Malignant) model with data pre-processing stage in MIAS dataset. where 1 = positive (abnormal), 0 = negative (normal)

Table 8 Comparison of Normal versus Cancer (Malignant only) and Normal versus Abnormal (benign and malignant) models with and without data pre-processing stage in INbreast dataset

Model	AUC	F1 score	Accuracy	Training time total (s)	Training time avg per epoch (s)	Avg predicting time per sample (s)
			Without data pre-processing			
Normal versus cancer	77.55%	71.43%	71.43%	4.32	0.07	0.05
Normal versus abnormal	65.30%	62.50%	57.14%	5.52	0.09	0.06
			With data pre-processing			
Normal versus cancer	94.25%	91.14%	91.84%	82.97	1.38	0.39
Normal versus abnormal	89.79%	87.17%	88.44%	82.78	1.38	0.42

example is from negative samples and the other two are from positive samples. In Normal versus Abnormal model, there are six examples from ten examples that have wrong predictions. From negative samples, there is one incorrect prediction. From positive samples, there are five incorrect predicted.

2. **MIAS Dataset.** This experiment evaluates the results of the Normal versus Cancer (Malignant only) model and the Normal versus Abnormal (Benign and Malignant) model in the MIAS dataset [32] without data pre-processing steps. Table 9 shows the results of both models with and without data pre-processing stage in terms of AUC, F1 score, accuracy, and time complexity.
   As shown in Table 9 also, in this dataset, the models have performance declines. The difference ratio between the Normal versus Cancer models with and without data pre-processing of AUC is 51.7%, the F1 score is 56.86%, and accuracy 41.76%, while the difference ratio between the Normal versus Abnormal models with and without data pre-processing of AUC is 24.38%, the F1 score is 9.1%, and accuracy 27.87%.
   Figures 20 and 21 present the predictions of some testing examples from both models. For Normal vs Cancer model, we can see that there are six examples from ten examples that have wrong predictions. From negative samples, there is one incorrect prediction. From positive samples, there are five incorrect predicted. For Normal versus Abnormal model, there are seven examples predicted wrong from ten examples. The six wrong examples are from negative samples, while only one wrong example is from positive samples.

In this section, we answered the second question of the research questions and get these inferences. The pre-processing of the data significantly affected the performance of the proposed framework. The data pre-processing stage is an essential step to get prominent performance.

5.3 Experiment 3: RQ3. How to Evaluate the Effectiveness of the Proposed Framework for Breast Cancer Diagnosis?

In this section, we will answer the third question of the research questions. The answers include two parts. The first part of the answer to the question of **how to evaluate a framework** is by using the evaluation metrics as in previous sections. The second part of the answer to the question of **how to evaluate the effectiveness of the proposed framework** which is the significant part by comparing the performance of the proposed framework with recent and relevant works to see if the proposed framework outperform the previous works or not. This is the actual answer to **how to evaluate the effectiveness of the framework**.

5.3.1 Experiment 3.1: Comparing the Results of the Proposed Framework with Recent and Relevant Works

This experiment aims to illustrate the effectiveness of the proposed framework by comparing them with relevant works. There are a few works in the field of anomaly detection techniques for breast cancer to the best of our knowledge up to now. We found that only **two works** related to our work used public datasets since most of them used private datasets and were not allowed to access them. These two works used INbreast and MIAS datasets [23, 32]; for this reason, we chose the same datasets used in the previous works to compare with them. For a fair comparison, the compared works use the same dataset we used (INbreast and MIAS datasets) and also used the anomaly detection technique. In addition, we compared the results with the same evaluation metrics used in these previous works.

1. **INbreast Dataset**
   - ***Normal versus Cancer (Malignant Only) Model.*** We compare the Normal versus Cancer (Malignant only) model with recent work to recognize breast cancer from normal data. Table 10 compares the proposed anomaly detection model (Normal versus Cancer model) with another anomaly detection model [28] that used INbreast dataset [23].
     The work that was compared in this case is the work of Wei et al. [28], and it was performed in 2018 and it used different feature extraction and classification method. Moreover, it used the image patches technique when train and testing their model.
     We consider the AUC metric to compare the efficiency of the model's performance since the previous work used only this metric. From the Table 10, we can see that the performance of our proposed model is 94.25% and achieves better performance than previous work [28] in terms of AUC. Furthermore, this work [28] decomposes the image into patches of much smaller size, i.e., 128×128, while in our work, we used the entire image and that is the best. In the end, the proposed model outperformance of the prior work increased by up to 10.25%.
   - ***Normal versus Abnormal (Benign and Malignant) model.*** We compare the Normal versus Abnormal (Benign and Malignant) model with recent work to recognize the two types of breast cancer (benign and malignant) from normal data. Table 11 compares the

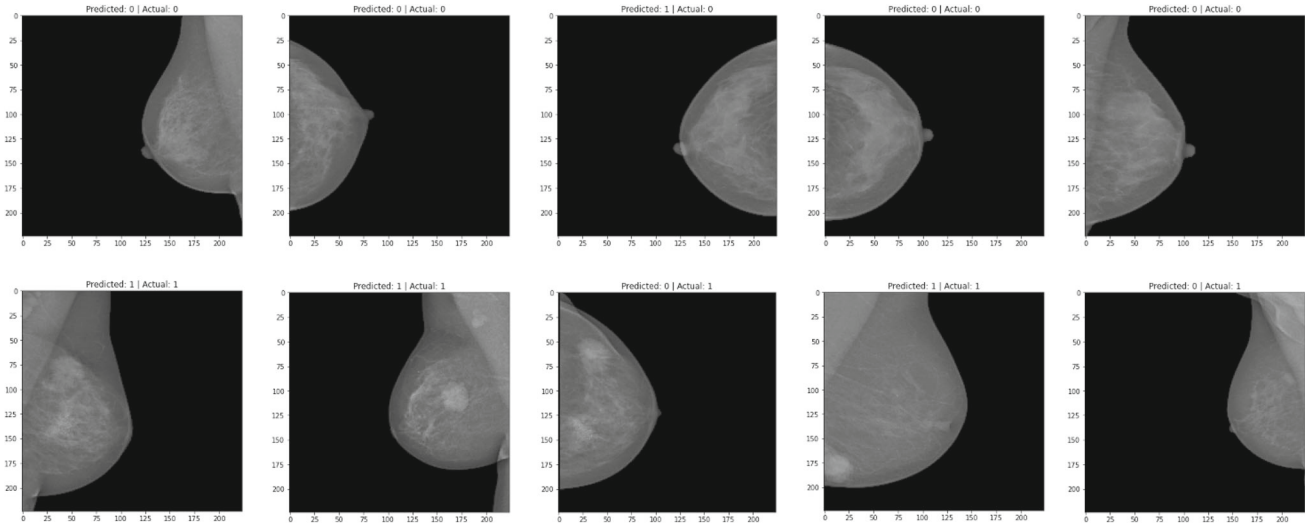


Fig. 18 Some testing examples from the Normal versus Cancer (Malignantonly) model without data pre-processing stage in INbreast dataset. Where 1 = Positive (cancer), 0 = negative (normal)

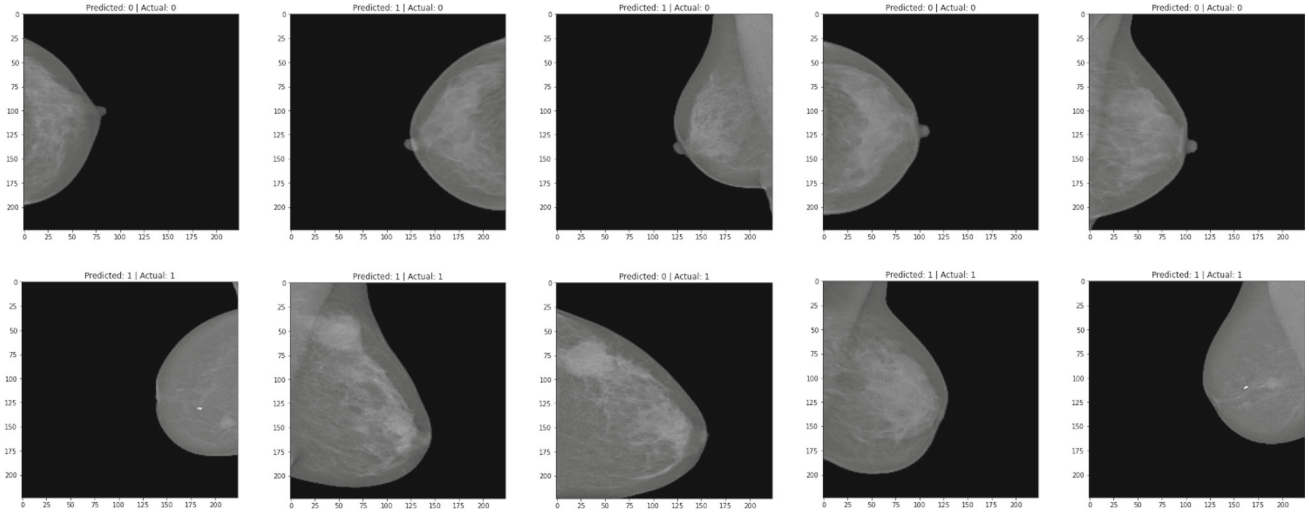


Fig. 19 Some testing examples from the Normal versus Abnormal (Benign and Malignant) model without data pre-processing stage in INbreast dataset, where 1 = positive (abnormal), 0 = negative (normal)

Table 9 Comparison of Normal versus Cancer (Malignant only) and Normal versus Abnormal (Benign and Malignant) models with and without data pre-processing stage in MIAS dataset

Model	AUC	F1 score	Accuracy	Training time total (s)	Training time avg per epoch (sec)	Avg predicting time per sample (s)
			Without data pre-processing			
Normal versus cancer	45.66%	35.29%	50.00%	8.94	0.15	0.09
Normal versus abnormal	57.02%	63.49%	47.73%	10.90	0.18	0.09
			With Data Pre-processing			
Normal versus cancer	97.36%	92.15%	91.76%	80.25	1.34	0.57
Normal versus abnormal	81.40%	72.59%	75.60%	83.26	1.39	0.56

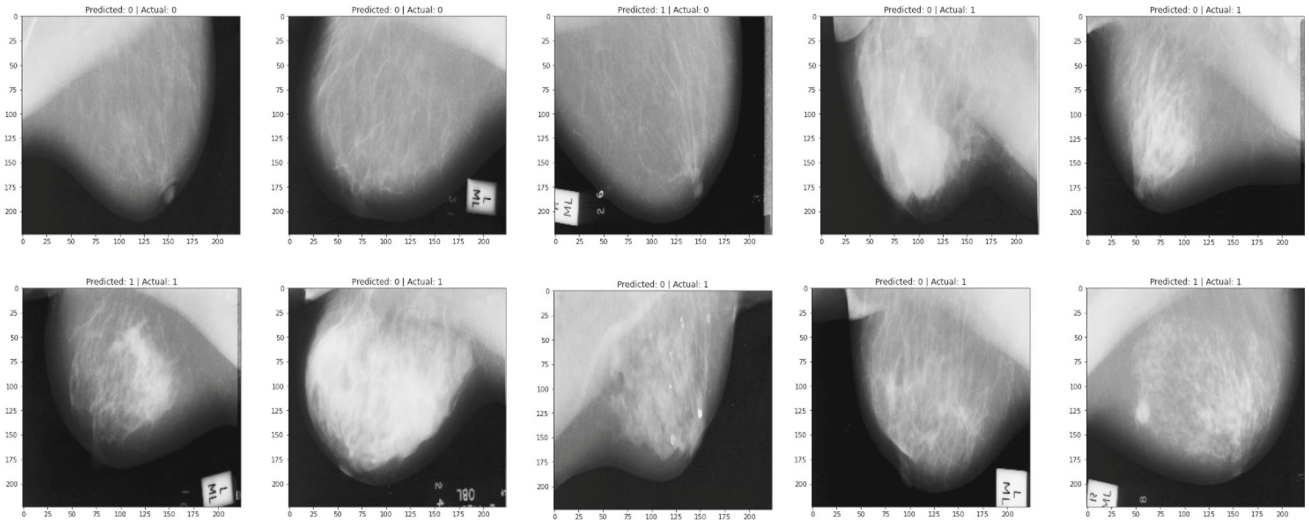


Fig. 20 Some testing examples from the Normal versus Cancer (Malignant only) model without data pre-processing stage in MIAS dataset, where 1 = positive (cancer), 0 = negative (normal)

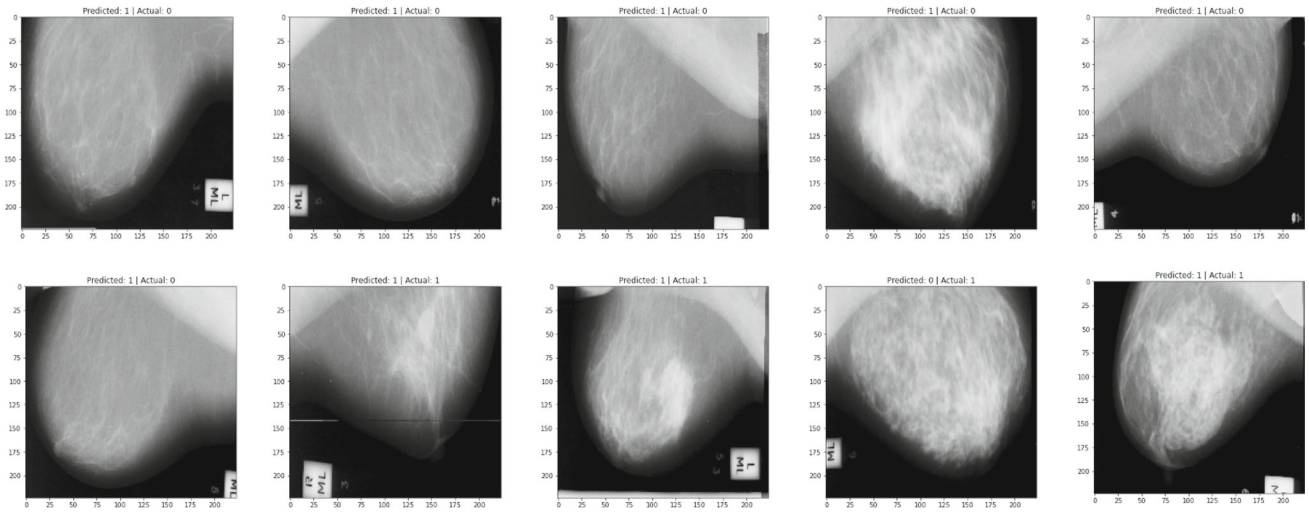


Fig. 21 Some testing examples from the Normal versus Abnormal (Benign and Malignant) model with data pre-processing stage in MIAS dataset, where 1 = positive (abnormal), 0 = negative (normal)

proposed anomaly detection model (Normal versus Abnormal model) with another anomaly detection model [10] that used INbreast dataset [23].

The work that was compared in this case is the work of Ahmadi et al. [10], and it was performed in 2019 and it used different feature extraction and classification method. Moreover, it also used the image patches technique when training and testing their model.

We consider the AUC and F1 score metrics to compare the efficiency of the model's performance since the previous work used only these two metrics. From Table 11, we can see that the performance of our proposed model is 89.79% in terms of AUC, and 87.17% in F1 score and achieves better performance than previous work [10] in both two metrics. Furthermore, this work [10] decomposes the image into patches of much smaller size, i.e., 64×64, 128×128, 256×256, while in our work, we used the entire image and that is the direction future work of this previous work [10]. In the end, the proposed model outperformance of the previous work increased by up to 2.79% and 7.17% in terms of AUC and F1 score, respectively, if we consider the highest results in the patch size.

2. **MIAS Dataset**

- ***Normal versus Cancer (Malignant Only) Model.*** No works are using the same dataset (MIAS dataset) to recognize breast cancer (malignant only) in the anomaly detection field to compare our results. However, it is an open area and it could be a future work.

Our performance is present in Table 7, and we left it open to other research in the communities to take it and do further work on it and compare with it. This would be a benchmark in this case.

- ***Normal versus Abnormal (Benign and Malignant) Model.*** We compare the Normal versus Abnormal (Benign and Malignant) model with recent work in the MIAS dataset [32]. Table 12 compares the proposed anomaly detection model (Normal versus Abnormal model) with another anomaly detection model [10].
  The work that compares in this case is the work of Ahmadi et al. [10], and it is the same work in the previous case (Normal versus Abnormal in INbreast dataset).
  From Table 12, we can see that the performance of our proposed model is 81.40% and 72.59%, and also in this case, our proposed model achieves better performance than previous work [10] in terms of AUC and F1 score, respectively. Furthermore, our work in this case also used the entire image rather than the patches image as in previous work [10]. In the end, the proposed model outperformance of the previous work increased by up to 5.40% and 1.59% in terms of AUC and F1 score, respectively, if we consider the highest results in a patch size.

In this section, we answered the third question of the research questions and got this inference. The proposed framework outperform the recent and relevant works in both datasets with high results. Moreover, we have one benchmark model—Normal versus Cancer (Malignant only) model in the MIAS dataset—that could help the communities do further work on it and compare with it.

6 Discussion

Figures 22, 23 and 24 summarize the results of the proposed framework, and the following was observed:

In the First Experiment: we notice that the Normal versus Cancer model results are better than the results of the Normal versus Abnormal model in INbreast dataset. The main reason is the model can easily distinguish between cancer (one case) versus normal. However, it is difficult to differentiate between the abnormal (more than one case and the benign case may close to a normal case) versus normal. In the end, the model has achieved good performance with this challenge. In MIAS dataset, the AUC of Normal versus Cancer model in the MIAS dataset is superior to the AUC of the same model in the INbreast dataset. The main reason is that the number of negative data with and without augmentation in this dataset is more than in the INbreast dataset. Although that the images of this dataset have a lower resolution and have noise, the number of the negative data and the model has one case to distinguish (malignant only) affected the result to increase it. Ultimately, this model has achieved excellent performance for this dataset. In addition, some reasons affected the results of Normal versus Abnormal model in MIAS dataset. First, the model can easily distinguish between cancer (one case) versus normal. However, it is difficult to differentiate between the abnormal (more than one case and the benign case may close to a normal case) versus normal; this is why the results decreased than the Normal versus Cancer model in the same dataset. Second, the images in this dataset have a lower resolution where the resolution is 1024×1024; however, in the INbreast dataset, the resolution is 3000×4000 approximately. Moreover, there is high noise in this dataset. Therefore, all these factors affected the results of this model. Ultimately, the model has achieved good performance with all these challenges.

In the Second Experiment: the Normal versus Abnormal model is lower than the Normal versus Cancer model in INbreast dataset since it has two different cases (benign and malignant) and a normal case to distinguish them. In the end, the model faced many difficulties, which led to a decrease in its performance. In MIAS dataset, the AUC of Normal versus Cancer model is much lower than the AUC of the same model in the INbreast dataset, while the opposite is in the presence of data pre-processing. The main reason is that the number of negative data became small in which augmentation technique was removed since it is in the data pre-processing stage. In addition, the images of this dataset have a lower resolution and have noise. Finally, the model has poor performance for this dataset.

In the Third Experiment: we find that the proposed framework outperforms the recent and relevant works in both datasets with prominent results. Furthermore, to the best of our knowledge up to now, this work is the first work that outperforms the previous works in the same datasets and the same technique.

As shown in Fig. 25, the time complexity of the proposed framework is acceptable according to the number of images used and the processing of them including the data pre-processing stage, feature extraction, and classifier used. But, the time complexity of the proposed framework without the data pre-processing stage is shortest since there are no pre-processing steps (see Fig. 26).

In the end, the proposed framework achieved prominent performance and became more reliable. We can claim that with this anomaly detection framework no need to gather a dataset containing all types of cancer cells and tissues as we might need in the case of building binary classifiers to get high performance [10].

Table 10 Comparison of results of Normal versus Cancer (malignant only) model with recent work on INbreast dataset [a]

Research paper	Patch size	AUC	Findings
Wei et al. [28] (2018)	128 × 128	84%	Image patches (patch-based method)
This work	**Entire image**	**94.25%**	**Whole image**

[a]The best performance is typeset in Bold

Table 11 Comparison of results of Normal versus Abnormal (Benign and Malignant) model with recent work on INbreast dataset [a]

Research paper	Patch size	AUC	F1 Score	Findings
Ahmadi et al. [10] (2019)	64	74%	69%	Image patches (patch-based method)
	128	87%	80%	
	256	86%	79%	
This work	**Entire image**	**89.79%**	**87.17%**	**Whole image**

[a]The best performance is typeset in Bold

Table 12 Comparison of results of Normal versus Abnormal (Benign and Malignant) model with recent work on MIAS dataset [a]

Research paper	Patch size	AUC	F1 Score	Findings
Ahmadi et al. [10] (2019)	64	72%	67%	Image patches(patch-based method)
	128	76%	71%	
	256	76%	70%	
This work	**Entire image**	**81.40%**	**72.59%**	**Whole image**

[a]The best performance is typeset in Bold

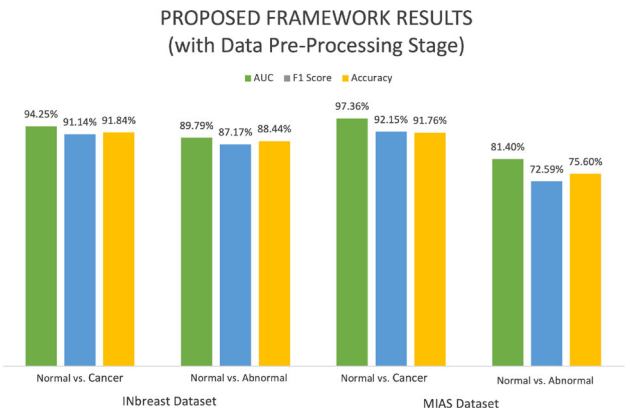


Fig. 22 The proposed framework results with data pre-processing stage

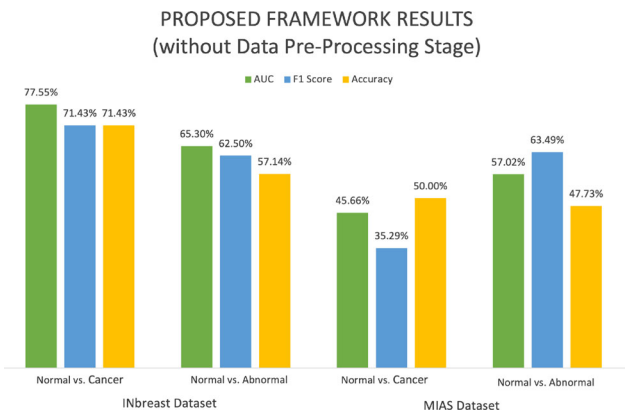


Fig. 23 The proposed framework results without data pre-processing stage

7 Threats to Validity

This section will present some threats to the validity of this work. It includes two subsections: internal validity threats and external validity threats.

7.1 Internal Validity Threats

Internal validity is the reliability of a study's cause-and-effect relationship between a treatment and a result [48, 49].

When using a small number of images or a small dataset size, since we cannot get more real images, especially in the medical field, the model will not be training well, and hence we get depressed performance. To avoid this, we used the augmentation technique to increase the images number or the dataset size, and then the model can learn better, as, if the training size increases, the accuracy can be improved. This technique improved the model's performance and allowed us to get prominent performance. However, this technique affects the results negatively too, since the performance will go down if the augmentation technique is not used and the original dataset size is used.

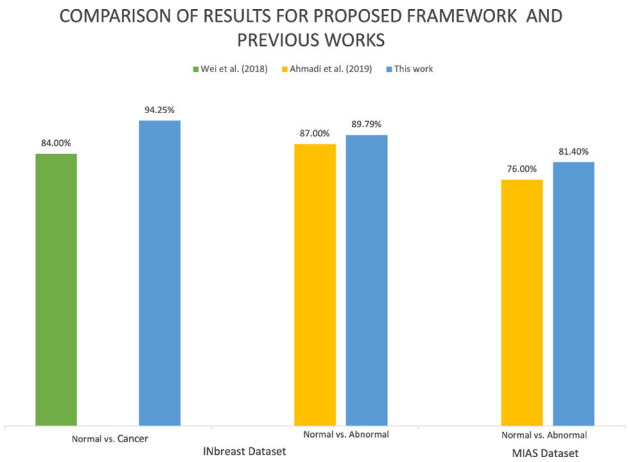


Fig. 24 The comparison of results of the proposed framework with recent works

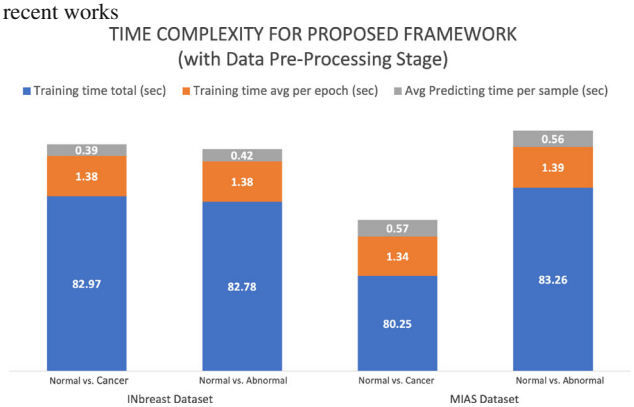


Fig. 25 The time complexity of proposed framework with data pre-processing stage

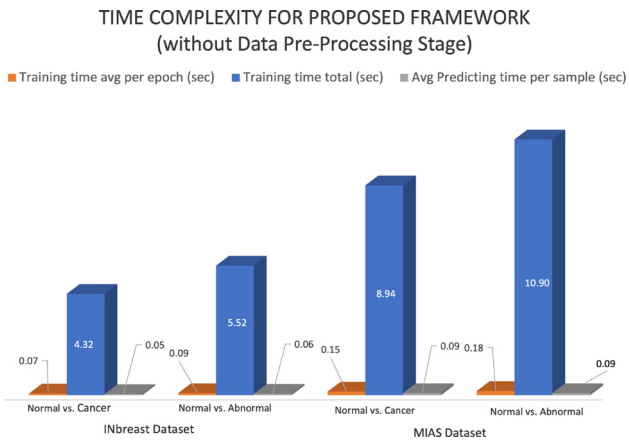


Fig. 26 The time complexity of proposed framework without data pre-processing stage

7.2 External Validity Threats

External validity is the extent to which the findings of a study can be generalized [48, 49].

We are interested in the generalization of the results. The obtained framework's performance was the best performing at the time of this study preparation. We did not evaluate the generalizability of the proposed framework. So this could be a limitation of this work; however, it could be defined as directions for our future works.

8 Conclusion and Future Work

In this article, we developed eight efficient and accurate anomaly detection models for breast cancer diagnosis using deep learning with two different datasets and different techniques. The primary goal is to address the problem of the imbalanced data issue that is popular in the medical field. Also, we aimed to minimize the effort required in detecting breast cancer manually and mitigate any error in the diagnosis.

The proposed framework contains two main stages: (1) data pre-processing and (2) features extraction and model development. The data pre-processing is an essential step in preparing the images for the features extraction stage. Also, it helps improve the performance significantly. It is important to note that each dataset has different pre-processing steps according to its properties. We used a MobileNetV2 pre-trained model for extracting deep features concerning the feature extraction and model development stage. The classifier network is a single-layer perceptron (SLP) that takes the features extracted by the feature extractor network and predicts whether the sample belongs to the normal class, cancer class, or abnormal class.

The experimental results have shown that the proposed framework in the INbreast dataset detects breast cancer from normal data with a high AUC rate of 94.25% and about a 10.25% increase in performance compared to recent relevant work. In addition, the framework detects breast abnormalities from normal data in the INbreast dataset with a high AUC rate of 89.79% and about a 2.79% increase in performance compared to recent relevant work. It is also reported that the framework detects breast cancer from normal data for the MIAS dataset with a high AUC rate of 97.36%, and the framework detects breast abnormalities from normal data in the MIAS dataset with a high AUC rate of 81.40% and about a 5.40% increase in performance compared to recent relevant work. In addition, the proposed framework has overcome the limitations of previous works since they implement the model in the way of image patches (patch-based method). The future work of these recent works was to expand the

model to process the whole image, implemented in this article.

As future work, the proposed framework can be extended as follows:

1. Apply the domain generalization techniques to the proposed framework for adaptability.
2. Apply the proposed framework using local data from Saudi Arabia.
3. Defining more classification classes to recognize the type of abnormal cases.
4. Comparison of our Normal versus Cancer (Malignant only) model in MIAS dataset with next future works of research communities.
5. Build a mobile application to operate the proposed framework on it.

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